This chapter will help you learn how to enable Linux binary compatibility on Unix-based FreeBSD by installing shared libraries and applications on the latter. Additionally, you will also learn about some implementation details that will help you develop binary compatible Linux software and applications. But before heading towards anything else, it is essential to understand what a binary compatible Linux OS means.

What is a Binary Compatible Linux: A binary compatible operating system, in this case, Linux, is one that focuses on implementing compatibility with other OSes or variants of the same OS. Since an OS's primary task is to run computer programs, it is essential to have the same Instruction Set Architecture (ISA) to make two OS B binary compatible. However, it is not a necessity. For instance, Linux Kernel and Windows are not compatible. You can still make Linux programs and apps that can run on Windows by making them compatible. This can be done using third-party applications such as Wine.

You will be learning more about Wine and how to use it in Linux in another chapter of this magazine. What is important for now is to know that you can employ third-party software in a CPU emulator to make Linux compatible with other OSes.

Making Linux binaries compatible with UNIX-based FreeBSD.

In addition to knowing what a binary compatible Linux is, it is essential to understand what it is not. An emulator can run programs made for guest OS, even if the host OS is not compatible. The same is the case with virtualisation, enabling the host OS to run programs made for guest OS. But